

es; inability to distinguish between human vs machine-controlled actions; misuse of the techniques during interrogation by the authorities; mind reading capability and privacy; and issues of mind control and emotion control. Fixation of legal responsibility in cases where accidents happen is an area of concern.

How would it be possible to distinguish a voluntary action from a malfunction of the system? Legal issues may also arise where the responses are obtained from seriously ill patients, as to whether these can be considered an informed consent or not given their status. The associated safeguards need to be in place before the technology is deployed and the ethical guidelines for research, development and dissemination need to be formed with global consensus.

Despite the exciting and dynamic future that the BCIs hold, one challenge is the lack of complete understanding of the brain. The human brain contains about 100 billion neurons with each having up to 10,000 connections to its surrounding neurons. The dynamics of the electrochemical interactions between the neurons and how these are translated into memories, behaviours, perceptions, and actions is still not fully understood.

The other challenge is the body's response to invasive BCIs using an implanted array of micro-electrodes in direct contact with the specific neurons in the brain. The electrodes are recognised as a foreign body triggering the natural immune system and they get surrounded by fibrous capsules of tissue, reducing the signal recording ability of the electrodes, and ultimately causing the BCIs to become unusable.

Reducing power consumption to decrease the battery size and prolong lifespan is another challenge as there is a trade-off between the power consumption and efficient bio-security. To ensure bio-security, the signals need to be encrypted, and this increases the power consumption.

The biggest impediment to the BCI technology is the lack of a sensor modality that provides safe, accurate, and robust access to brain signals.



Fig. 10: Elon Musk's Neuralink brain chip (Source: www.youtube.com)

Also developing sensors with additional channels to improve accuracy and minimising their power usage is a challenge. The ethical, legal, and social implications of BCIs may slow down, stop, or divert the technology into something completely different from what was aimed.

Conclusion and the future

All the information inside the brain cannot be put in a linguistic form, as it is difficult to verbalise it into words. For instance, an experienced surgeon cannot simply 'tell' an intern how to exactly position and move the fingers during a critical surgery. BCIs provide alternate methods to interact with the outside world as they do not depend on the brain's normal output pathways such as peripheral nerves and muscles. Direct brain-to-brain communication has been driven by the motives as diverse as futurist enthusiasm and military exigency, to make it possible to replace language by directly linking the activity of human brains and ultimately providing a new means of communication.

However, despite all that the scientists and researchers have achieved in recent times, the mind remains a black box. To open a tiny crack in that black box is a significant achievement as there are many unknown areas such as interpreting the brain electric activity associated with deeper experiences like hope, love, or sadness.

In the last few years, BCI research has gathered significant momentum. Mark Zuckerberg recently revealed that they are looking at the brain as the next big communication platform and hope to produce a BCI that will allow people to type three times faster.

Elon Musk has created Neura-6 link to provide a direct cortical interface with the brain via neural lace, to help humanity better compete with AI. He is developing a robotically implantable BCI containing 3,000 electrodes to provide extensive interaction between computers and the brain nerve cells.

Kernal, headed by Bryan Johnson, is going big with a BCI device to enable people to learn faster, remember more,

'co-evolve' with artificial intelligence, and even connect into group minds. Charles Stark Draper Labs is developing wirelessly connected implants for prosthetics, a live remote-controlled dragonfly with a BCI, and a field cap to read neural activity noninvasively. DARPA is developing an implantable neural interface capable of engaging one million nerve cells simultaneously.

Brain-to-brain communication has potential for use in a future battlefield to enhance the speed of communication, improve situational awareness, and allow operators to control multiple technological platforms simultaneously. Neurotechnological advances have given quadriplegics the ability to perform in an F-35 simulator with their thoughts and the scientists the ability to decode the speech that the subjects are imagining, albeit imperfectly.

Nicolelis, a Brazilian soccer aficionado, built a robotic exoskeleton controlled by EEG impulses for the 2014 World Cup to enable a young paraplegic person to deliver the ceremonial first kick.

In the last decade, the genetic sequencing technology and modern tools for mapping have helped in increased understanding of the neuronal firing patterns and how they lead to different actions. Brain-interfacing devices are becoming more sensitive, smaller, smarter, and portable over time. Future technologies need to address issues related to ease of use, performance robustness, and cost reduction. **EFY**



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